Statistical Modelling & Analysis (STAT2004)
Applied Class (Week 3)

We will go through Question 1, 2 & 5 together in the Applied Class. Try the other questions yourself during the week and discuss your results with your tutor next week.

1. The number of trials required to obtain the first success is modelled by the geometric distribution, which has pmf

$$f(x;p) = p(1-p)^{x-1}, \qquad x = 1, 2, 3, \ldots \qquad p \in (0, 1].$$

Suppose $X_1, \ldots, X_n$ is a random sample from this distribution.

(a) Write down the likelihood function for this model.

(b) Write down the score function.

(c) Determine the maximum likelihood estimator of p. Make sure you demonstrate that you have found the global maximum of the likelihood, including the possibility that the maximum occurs at the boundary $p = 1$.

Solution. **(a)** Let

$$S = \sum_{i=1}^{n} x_i.$$

The likelihood is

$$\begin{aligned} L(p) &= \prod_{i=1}^{n} p(1-p)^{x_i - 1} \\ &= p^n (1-p)^{S-n}, \qquad 0 < p \leq 1. \end{aligned}$$

(b) The log-likelihood is

$$\ell(p) = n \log p + (S - n) \log(1 - p).$$

For an interior point $0 < p < 1$, the score function is

$$U(p) = \frac{d\ell(p)}{dp} = \frac{n}{p} - \frac{S-n}{1-p}.$$

(c) First suppose $S > n$. Setting the score equal to zero gives

$$\frac{n}{p} = \frac{S-n}{1-p}.$$

Therefore

$$n(1-p) = p(S-n),$$

and hence

$$n = pS.$$

Thus

$$\hat{p} = \frac{n}{S} = \frac{1}{\bar{X}}.$$

To check that this gives the global maximum when $S > n$, observe that

$$\ell'(p) = \frac{n}{p} - \frac{S-n}{1-p}.$$

The derivative is positive when $p < n/S$ and negative when $p > n/S$. Thus the log-likelihood increases up to $p = n/S$ and decreases after $p = n/S$, so $p = n/S$ is the global maximum.

If $S = n$, then every observation is equal to one. In this case

$$L(p) = p^n, \qquad 0 < p \leq 1,$$

which is maximised at the boundary $p = 1$. The formula $\hat{p} = n/S$ also gives $\hat{p} = 1$ in this case. Therefore, in all cases,

$$\hat{p} = \frac{n}{\sum_{i=1}^n X_i} = \frac{1}{\bar{X}}.$$

2. Suppose $X_1, \ldots, X_n$ is a random sample from a distribution with probability density function

$$f(x;\theta) = \tfrac{1}{2}(1 + \theta x), \qquad x \in (-1, 1), \qquad \theta \in [-1, 1].$$

(a) Write down the likelihood function for this model.

(b) Write down the score function.

(c) Show that the MLE exists. Also show that, except for a case which occurs with probability zero, the MLE is unique.

(d) Explain why the MLE may occur at the boundary $\theta = -1$ or $\theta = 1$.

Solution. **(a)** The likelihood is

$$L(\theta) = \prod_{i=1}^n \frac{1}{2}(1 + \theta x_i) = 2^{-n} \prod_{i=1}^n (1 + \theta x_i), \qquad -1 \leq \theta \leq 1.$$

(b) The log-likelihood is

$$\ell(\theta) = -n \log 2 + \sum_{i=1}^n \log(1 + \theta x_i).$$

Hence the score function, for $-1 < \theta < 1$, is

$$U(\theta) = \frac{d\ell(\theta)}{d\theta} = \sum_{i=1}^n \frac{x_i}{1 + \theta x_i}.$$

(c) Since $x_i \in (-1, 1)$ and $\theta \in [-1, 1]$, we have

$$1 + \theta x_i > 0.$$

Hence the likelihood is continuous on the closed interval $[-1, 1]$. A continuous function on a closed bounded interval attains a maximum, so the MLE exists.
To discuss uniqueness, differentiate the score:

$$U'(\theta) = -\sum_{i=1}^n \frac{x_i^2}{(1 + \theta x_i)^2}.$$

Except in the case $x_1 = \cdots = x_n = 0$, this derivative is strictly negative. Thus, except for this case, the score $U(\theta)$ is strictly decreasing as a function of θ.
Since the observations come from a continuous distribution, the probability that all observations are exactly zero is zero. Therefore, except for a case that occurs with probability zero, the score is strictly decreasing. It can cross zero at most once. If it crosses zero, there is a unique interior maximiser. If it does not cross zero, the maximum occurs at one of the two endpoints. In either case, except for the probability-zero case, the MLE is unique.

(d) The MLE may occur at the boundary because the largest value of the likelihood over $[-1, 1]$ need not occur at a point where the score is zero. In particular, if all $x_i > 0$, then the derivative of the log likelihood (ie the score function) is

$$U(\theta) = \frac{d\ell(\theta)}{d\theta} = \sum_{i=1}^{n} \frac{x_i}{1 + \theta x_i} > 0 \qquad \text{for all } \theta \in [-1, 1].$$

In this case $\hat{\theta} = 1$. Similarly, if all $x_i < 0$, then $\hat{\theta} = -1$.

3. Suppose $X_1, \ldots, X_n$ is a random sample from $\mathsf{U}[\theta, 2\theta]$, that is the uniform distribution on $[\theta, 2\theta]$, where $\theta > 0$.
   (a) Write down the likelihood function $\mathcal{L}(\theta)$ for this model.
   (b) Determine maximum likelihood estimator (MLE). (Remember not every MLE is found by differentiating.)
   (c) Show that the MLE is consistent. It may help to recall that if $X_1, \ldots, X_n$ are independent and $Y = \max\{X_1, \ldots, X_n\}$, then

$$\mathbb{P}(Y < c) = \mathbb{P}(X_1 < c, \ldots, X_n < c) = \prod_{i=1}^{n} \mathbb{P}(X_i < c), \qquad \text{for any } c > 0.$$

   (d) Determine the bias of the MLE. It may help to first find the distribution function of $X_{(n)}$.
   (e) Determine the mean squared error of the MLE.

Solution. (a) Let

$$x_{(1)} = \min\{x_1, \ldots, x_n\}, \qquad x_{(n)} = \max\{x_1, \ldots, x_n\}.$$

The density of one observation is

$$f(x; \theta) = \frac{1}{\theta} \mathbb{I}(\theta \le x \le 2\theta).$$

Hence the likelihood is

$$\mathcal{L}(\theta) = \theta^{-n} \mathbb{I}\{\theta \le x_{(1)},\ x_{(n)} \le 2\theta\}.$$

Equivalently,

$$\mathcal{L}(\theta) = \theta^{-n} \mathbb{I}\left\{\frac{x_{(n)}}{2} \le \theta \le x_{(1)}\right\}.$$

(b) The likelihood is nonzero only when

$$\frac{x_{(n)}}{2} \le \theta \le x_{(1)}.$$

On this interval, θ^{-n} decreases as θ increases. Therefore the likelihood is maximised by choosing the smallest allowable value of θ. Hence

$$\hat{\theta} = \frac{X_{(n)}}{2}.$$

(c) Since $X_{(n)} \le 2\theta$, we have $\hat{\theta} \le \theta$. For $0 < \epsilon < \theta/2$,

$$\begin{aligned} \mathbb{P}_\theta(|\hat{\theta} - \theta| > \epsilon) &= \mathbb{P}_\theta\left(\theta - \frac{X_{(n)}}{2} > \epsilon\right) \\ &= \mathbb{P}_\theta(X_{(n)} < 2\theta - 2\epsilon). \end{aligned}$$

For $\theta \le c \le 2\theta$,

$$\mathbb{P}_\theta(X_i < c) = \frac{c-\theta}{\theta}.$$

Thus, with $c = 2\theta - 2\epsilon$,

$$\mathbb{P}_\theta(X_{(n)} < 2\theta - 2\epsilon) = \left(1 - \frac{2\epsilon}{\theta}\right)^n.$$

This tends to zero as $n \to \infty$. Hence $\hat{\theta} \to \theta$ in probability, so the MLE is consistent.

(d) For $\theta \le y \le 2\theta$,

$$\begin{aligned}\mathbb{P}_\theta(X_{(n)} \le y) &= \prod_{i=1}^{n} \mathbb{P}_\theta(X_i \le y) \\ &= \left(\frac{y-\theta}{\theta}\right)^n.\end{aligned}$$

Define

$$W = \frac{X_{(n)} - \theta}{\theta}.$$

Then $0 \le W \le 1$ and

$$\mathbb{P}(W \le w) = w^n, \qquad 0 \le w \le 1.$$

Therefore W has density nw^{n-1} on $(0,1)$, and

$$\mathbb{E}(W) = \int_0^1 w n w^{n-1}\, dw = \frac{n}{n+1}.$$

Hence

$$\mathbb{E}_\theta(X_{(n)}) = \theta + \theta\mathbb{E}(W) = \theta\left(1 + \frac{n}{n+1}\right) = \theta\frac{2n+1}{n+1}.$$

Thus

$$\mathbb{E}_\theta(\hat{\theta}) = \frac{1}{2}\mathbb{E}_\theta(X_{(n)}) = \theta\frac{2n+1}{2(n+1)}.$$

Therefore

$$\mathrm{Bias}(\hat{\theta}) = \mathbb{E}_\theta(\hat{\theta}) - \theta = -\frac{\theta}{2(n+1)}.$$

The MLE is biased downward.

(e) We use

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2.$$

From the density of W above,

$$\mathbb{E}(W^2) = \int_0^1 w^2 n w^{n-1}\, dw = \frac{n}{n+2}.$$

Therefore

$$\begin{aligned}\mathrm{Var}(W) &= \frac{n}{n+2} - \left(\frac{n}{n+1}\right)^2 \\ &= \frac{n}{(n+1)^2(n+2)}.\end{aligned}$$

Since $X_{(n)} = \theta(1+W)$,

$$\mathrm{Var}(X_{(n)}) = \theta^2\,\mathrm{Var}(W) = \frac{n\theta^2}{(n+1)^2(n+2)}.$$

Hence

$$\mathrm{Var}(\hat{\theta}) = \frac{1}{4}\,\mathrm{Var}(X_{(n)}) = \frac{n\theta^2}{4(n+1)^2(n+2)}.$$

Also,

$$\mathrm{Bias}(\hat{\theta})^2 = \frac{\theta^2}{4(n+1)^2}.$$

Therefore

$$\begin{aligned}\mathrm{MSE}(\hat{\theta}) &= \frac{n\theta^2}{4(n+1)^2(n+2)} + \frac{\theta^2}{4(n+1)^2} \\ &= \frac{\theta^2}{2(n+1)(n+2)}.\end{aligned}$$

4. Suppose $X_1, \ldots, X_n$ is a random sample from $\mathsf{N}(\mu, 1)$.
   (a) Determine the MLE of $\exp(\mu)$. (You can use what you know about the MLE of μ.)
   (b) Is the MLE of $\exp(\mu)$ an unbiased estimator? It may help to recall the moment generating function of $Y \sim \mathsf{N}(\mu, \sigma^2)$ is $M(t) = \mathbb{E}(\exp(tY)) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2),\ t \in \mathbb{R}$.
   (c) Construct an unbiased estimator of $\exp(\mu)$ base on the MLE.

   Solution. **(a)** The MLE of μ is

$$\hat{\mu} = \bar{X}.$$

   By the invariance property of maximum likelihood estimators, the MLE of $\exp(\mu)$ is

$$\widehat{\exp(\mu)}_{\mathrm{MLE}} = \exp(\hat{\mu}) = \exp(\bar{X}).$$

   (b) Since $X_i \sim \mathsf{N}(\mu, 1)$,

$$\bar{X} \sim \mathsf{N}\left(\mu, \frac{1}{n}\right).$$

   Using the moment generating function of a normal random variable with $t = 1$,

$$\mathbb{E}_\mu\{\exp(\bar{X})\} = \exp\left(\mu + \frac{1}{2n}\right).$$

   This is not equal to $\exp(\mu)$. Therefore $\exp(\bar{X})$ is not an unbiased estimator of $\exp(\mu)$. Its bias is

$$\mathrm{Bias}\{\exp(\bar{X})\} = \exp(\mu)\left\{\exp\left(\frac{1}{2n}\right) - 1\right\}.$$

   (c) From part (b),

$$\mathbb{E}_\mu\{\exp(\bar{X})\} = \exp(\mu)\exp\left(\frac{1}{2n}\right).$$

   Hence multiplying the MLE by $\exp\{-1/(2n)\}$ removes the bias. Thus

$$\exp\left(\bar{X} - \frac{1}{2n}\right)$$

   is an unbiased estimator of $\exp(\mu)$.

5. The zero inflated Poisson distribution has probability mass function

$$f(x; \pi, \lambda) = \begin{cases} \pi + (1-\pi)\exp(-\lambda) & x = 0 \\ (1-\pi)\exp(-\lambda)\frac{\lambda^x}{x!} & x = 1, 2, \ldots \end{cases}$$

where $0 \leq \pi \leq 1$ and $\lambda > 0$. This distribution is used to model count data with more zeros than would be expected under the usual Poisson distribution.

The file `fish.csv` contains data on groups visiting a state park. The variable `count` records the number of fish caught by each group. The data contain many zeros, partly because some groups did not fish and partly because some groups fished but caught no fish. We will only consider the counts which can be obtained using the code:

```
fish <- read.csv("https://stats.idre.ucla.edu/stat/data/fish.csv")
data <- fish$count
```

(a) Write down the log likelihood for this model.

(b) In R or Python, write a suitable function to evaluate the log-likelihood for this model. Use `optim` in R (or an equivalent numerical optimiser in Python) to find the maximum likelihood estimate for this data. You may want to use the method of moments estimator as an initial value. (The method of moments estimator for this distribution was discussed in the Lecture 01, bottom of page 6 to page 7.)

Solution. **(a)** Let $x_1, \ldots, x_n$ denote the observed counts. The log-likelihood is

$$\ell(\pi, \lambda) = \sum_{i:x_i=0} \log\{\pi + (1-\pi)\exp(-\lambda)\} + \sum_{i:x_i>0} \left[\log(1-\pi) - \lambda + x_i \log\lambda - \log(x_i!)\right].$$

The first sum is the contribution from the zero observations. The second sum is the contribution from the positive observations.

(b) The following R code evaluates the negative log-likelihood using the original parameters (π, λ) and then minimises it using `optim` with method `L-BFGS-B`. The lower and upper bounds enforce $0 \leq \pi \leq 1$ and $\lambda > 0$, up to a small numerical tolerance.

```
fish <- read.csv("https://stats.idre.ucla.edu/stat/data/fish.csv")
data <- fish$count

# negative log likelihood

zip_negloglik <- function(theta, data){
  p <- theta[1]
  lambda <- theta[2]

  loglik <- rep(0, length(data))
  zeros <- (data==0)
  loglik[-zeros] <- log(1-p)  -lambda + data[-zeros]*log(lambda) +
                                        lfactorial(data[-zeros])
  loglik[zeros] <- log(p + (1-p)*exp(-lambda))
  return(-sum(loglik))
}

# Method of moment estimates

m1 <- mean(data)
m2 <- mean(data^2)

lambda_hat <- m2/m1 - 1
```

```
pi_hat <- 1 - m1/lambda_hat


# Find the MLE

theta0 <- c(pi_hat, lambda_hat)
theta_MLE <- optim(theta0,
                   zip_negloglik,
                   method = "L-BFGS-B",
                   lower = c(1e-8,1e-8),
                   upper = c(1 - 1e-8, Inf),
                   data = data)

theta_MLE
```

Printing `theta_MLE`

```
> theta_MLE
$par
[1] 0.567789 7.625909

$value
[1] -2485.994

$counts
function gradient
      19       19

$convergence
[1] 0

$message
[1] "CONVERGENCE: REL_REDUCTION_OF_F <= FACTR*EPSMCH"
```

The convergence code `$convergence` is 0 so `optim` was successful. `$message` tells us that no further improvement in the negative log likelihood was possible. The first component of `fit$par` is the MLE of π, and the second component is the MLE of λ.